Hoang Nguyen

hoangh@mit.edu | hoang-tn.github.io
Cambridge, Massachusetts, USA

Education

Massachusetts Institute of Technology (MIT), Cambridge, MA, USA 2024–Present
Ph.D., Electrical Engineering and Computer Science
Advisor: Priya L. Donti

Chung-Ang University (CAU), Seoul, South Korea 2021–2023
M.E., Electrical and Electronics Engineering
Graduated with Dean's Award

Hanoi University of Science and Technology (HUST), Hanoi, Vietnam 2015–2020
B.E., Control Engineering and Automation
Valedictorian – Rank: 1/297

Selected Awards & Honors

Grid Science Winter School and Conference Scholarship, Los Alamos National Laboratory 2027
Jin Au Kong Fellowship, MIT 2024
CAU Graduate School Dean's Award (for one outstanding Master's student in EEE Department) 2023
CAU Young Scientist Scholarship 2021–2023
Valedictorian Award, HUST 2020
Undergraduate Scholarships and Awards 2015–2020
Vallet Scholarship (France), IHI Scholarship (Japan), Nitori Scholarship (Japan), Best Student Awards

Research Experience

Laboratory for Information and Decision Systems, MIT 2024–Present
Research Assistant, advised by Prof. Priya Donti
- Developed a self-supervised learning method for constrained optimization with feasibility guarantees and order-of-magnitude speedups.
- Designed a model architecture for learning to solve constrained optimization that generalizes to unseen problem sizes 10–20× larger than the training instances.
- Proposed a scalable, robust learning framework for non-convex AC optimal power flow on thousand-node distribution grids, achieving constraint satisfaction, a negligible optimality gap, and an order-of-magnitude speedup.

Machine Decision Intelligence and Learning Lab, KAIST 2023–2024
Research Scientist, advised by Prof. Donghwan Lee
- Developed a sample-efficient safe reinforcement learning algorithm via successive convex approximation, with closed-form subproblem solutions that accelerates the training process.

Electric Power System and Smart Grid Lab, CAU 2021–2023
Research Assistant, advised by Prof. Dae-Hyun Choi
- Developed safe reinforcement learning methods for voltage control in distribution systems.
- Proposed distributionally robust MPC for EV charging and distribution operations, lowering daily cost while preserving reliability.

Department of Automatic Control, HUST 2017–2020
Research Assistant, advised by Prof. Nga Thi-Thuy Vu & Prof. Phuong Nam Dao
- Developed on- and off-policy Q-learning algorithms that solve the periodic LQR problem.
- Designed a robust controller for wind turbine generators with improved tracking performance over conventional controllers.

Publications

[1] **H. T. Nguyen** and P. Donti, "FSNet: Feasibility-seeking neural network for constrained optimization with guarantees," *Advances in neural information processing systems*, vol. 38, pp. 39 670–39 708, 2026, (featured on MIT News).

[2] **H. T. Nguyen**, S. Liu, R. S. Biswas, V. Pendyala, N. Virani, D. Deka, and P. L. Donti, "Scalable self-supervised learning for multiphase AC-OPF in distribution systems with topology reconfiguration," *arXiv preprint arXiv:2608.25095*, 2026.

[3] **H. T. Nguyen** and D.-H. Choi, "Distributionally Robust Model Predictive Control for Smart Electric Vehicle Charging Station with V2G/V2V Capability," *IEEE Transactions on Smart Grid*, vol. 14, no. 6, pp. 4621–4633, 2023.

[4] **H. T. Nguyen** and D.-H. Choi, "Distributionally Robust Safety Filter for Learning-Based Control in Active Distribution Systems," *IEEE Transactions on Smart Grid*, vol. 14, no. 6, pp. 4972–4975, 2023.

[5] **H. T. Nguyen**, Y.-J. Kim, and D.-H. Choi, "Sample-Efficient Learning for a Surrogate Model of Three-Phase Distribution System," *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 2361–2364, 2023.

[6] S. Jeon, **H. T. Nguyen**, and D.-H. Choi, "Safety-Integrated Online Deep Reinforcement Learning for Mobile Energy Storage System Scheduling and Volt/VAR Control in Power Distribution Networks," *IEEE Access*, vol. 11, pp. 34 440–34 455, 2023.

[7] **H. T. Nguyen** and D.-H. Choi, "Decentralized Distributionally Robust Coordination Between Distribution System and Charging Station Operators in Unbalanced Distribution Systems," *IEEE Transactions on Smart Grid*, vol. 14, no. 3, pp. 2164–2177, 2022.

[8] **H. T. Nguyen** and D.-H. Choi, "Three-Stage Inverter-Based Peak Shaving and Volt-VAR Control in Active Distribution Networks Using Online Safe Deep Reinforcement Learning," *IEEE Transactions on Smart Grid*, vol. 13, no. 4, pp. 3266–3277, 2022.

[9] **H. T. Nguyen**, H. Q. Nguyen, P. N. Dao, and N. T.-T. Vu, "On Robust Control of Permanent Magnet Synchronous Generators Using Robust Integral of Error Sign," in *Computational Intelligence Methods for Green Technology and Sustainable Development: Proceedings of the International Conference GTSD2020 5*, Springer, 2021, pp. 156–166.

[10] A. D. Hoang, H. Q. Nguyen, P. N. Dao, and **H. T. Nguyen**, "Output Feedback Adaptive Reinforcement Learning Tracking Control for Wheel Inverted Pendulum Systems," in *Research in Intelligent and Computing in Engineering: Select Proceedings of RICE 2020*, Springer, 2021, pp. 535–547.

[11] N. T. Dang, **H. T. Nguyen**, and P. N. Dao, "Modified Q-Learning Algorithm with Lifting Method for Discrete-Time Linear Periodic Systems," in *International Conference on Engineering Research and Applications*, Springer International Publishing Cham, 2020, pp. 548–559.

Posters and Presentations

2026 INFORMS Annual Meeting Nov. 2026
Scalable Self-supervised Learning for AC-OPF in Distribution Systems

5th Workshop on Foundation Models of the Electric Grid Mar. 2026
Self-supervised Learning for AC Optimal Power Flow in Distribution Systems with Hard Constraints

MIT-GE Vernova Energy and Climate Symposium Feb. 2026
AI for Large-Scale, Topology-Aware AC-OPF in Distribution Grids

39th Annual Conference on Neural Information Processing Systems Dec. 2025
FSNet: Feasibility-Seeking Neural Network for Constrained Optimization with Guarantees

Academic Service

Reviewer, IEEE Transactions on Smart Grid, IEEE Transactions on Power Systems, IEEE Transactions on Sustainable Energy — 2023–Present

Technical Skills

Programming: Python, Julia, MATLAB
Optimization: Gurobi, CPLEX, HiGHS, CVXPY, JuMP
Machine Learning: PyTorch, JAX, vLLM, SGLang
Spoken Languages: English (fluent), Vietnamese (native)